

Blooming Barrels Finance & CRM PWA: Database Architecture

Overview

This document outlines the database schema for the Blooming Barrels hybrid PWA, designed using PostgreSQL (ideal for Supabase or similar hosting). The architecture bridges personal finance tracking with a business CRM, supporting both manual entries and public link orders.

Table Definitions

1. Customers Table (CRM Core)

Stores data for Blooming Barrels clients to power marketing insights and birthday promos.

Column Name	Data Type	Constraints / Description
customer_id	UUID	Primary Key, Default Gen Random
name	VARCHAR(100)	Not Null
phone_number	VARCHAR(15)	Unique (Optional: useful for identifying repeats)
location_city	VARCHAR(50)	Not Null (Crucial for FB/TikTok ad targeting)
birthdate	DATE	Optional (For birthday promos)
created_at	TIMESTAMP	Default CURRENT_TIMESTAMP

2. Items Table (Inventory/Pre-defined Selections)

Stores the products sold (e.g., Cactus pups, Rooting Hormones) for quick selection during billing.

Column Name	Data Type	Constraints / Description
item_id	UUID	Primary Key, Default Gen Random
item_name	VARCHAR(100)	Not Null
default_price	DECIMAL(10, 2)	Not Null
category	VARCHAR(50)	E.g., 'Plants', 'Supplies'

3. Bills / Invoices Table (Business Revenue)

Records completed sales. When a record is inserted here, a trigger or backend logic should automatically log an entry in the Transactions table.

Column Name	Data Type	Constraints / Description
bill_id	UUID	Primary Key, Default Gen Random

Column Name	Data Type	Constraints / Description
customer_id	UUID	Foreign Key referencing Customers(customer_id)
total_amount	DECIMAL(12, 2)	Not Null
status	VARCHAR(20)	'Pending' (from link) or 'Completed' (billed)
payment_method	VARCHAR(20)	'COD', 'Bank Transfer', 'Cash'
created_at	TIMESTAMP	Default CURRENT_TIMESTAMP

4. Bill_Items Table (Junction Table)

Resolves the many-to-many relationship between Bills and Items (one bill can have many items).

Column Name	Data Type	Constraints / Description
bill_item_id	UUID	Primary Key, Default Gen Random
bill_id	UUID	Foreign Key referencing Bills(bill_id)
item_id	UUID	Foreign Key referencing Items(item_id)
quantity	INTEGER	Not Null, Default 1
unit_price	DECIMAL(10, 2)	Price at time of sale (in case default changes)

5. Transactions Table (Personal Finance Core)

The central ledger for all cash flow. Business sales flow here automatically; personal expenses and other incomes (e.g., Scholarships, CDS) are entered manually.

Column Name	Data Type	Constraints / Description
transaction_id	UUID	Primary Key, Default Gen Random
type	VARCHAR(10)	'Income' or 'Expense'
amount	DECIMAL(12, 2)	Not Null
category	VARCHAR(50)	'Blooming Barrels', 'Scholarship', 'Food', 'CDS' etc.
description	TEXT	Optional details or voice log text
linked_bill_id	UUID	Foreign Key referencing Bills(bill_id), Nullable
date	TIMESTAMP	Default CURRENT_TIMESTAMP

Key Logic Connections (For FastAPI Backend)

- **Auto-Income Sync:** When a new Bill is created with status 'Completed', the backend must create a new record in the Transactions table with type='Income', category='Blooming Barrels', amount=bill.total_amount, and linked_bill_id=bill.bill_id.
- **Public Link Flow:** Data coming from the public React form creates a Bill with status

'Pending' and creates/updates the Customer record. The Admin reviews this on the dashboard, changes status to 'Completed', triggering the Auto-Income Sync.